Response to Reviewers

Auditable agentic AI for evidence-grounded thyroid ultrasound diagnosis and reporting

Note. This draft contains the responses on report generation and its evaluation: Reviewer 1, Comments 3–6; Reviewer 2, Comments 1, 16–19 and 22; Reviewer 3, Major Comments 3 and 5 and Minor Comment 1. Reviewer 1, Comments 3 and 4, Reviewer 2, Comment 1, and Reviewer 3, Major Comment 3 are shared with the image-analysis part and will be merged with it. Numbers of new Supplementary Tables (from S12) and Figures (from S8) are provisional and continue the Supplementary Information of the submitted manuscript; they will be finalised when the revised Supplementary Information is assembled. Text in **[PENDING: …]** will be written when the clinician data or the image-analysis parts are available; **[AUTHOR: …]** marks information to be supplied by the authors.

Dear Editor and Reviewers,

We thank you for your careful evaluation of our manuscript, *Auditable agentic AI for evidence-grounded thyroid ultrasound diagnosis and reporting*, and for your constructive comments. We respond to each comment below. The reviewers' comments are shown in black italics and our responses in blue. New Supplementary Tables and Figures cited in the responses are reproduced at the end of this letter.

Response to Reviewer 1

Reviewer 1, Comment 3

Heterogeneity in external performance. *Although aggregate performance on the private multicentre cohort is encouraging, there is substantial centre-level variation. For example, nodule segmentation performance is much lower at some centres. The authors should investigate associations with scanner type, image quality, case mix, and annotation protocol, and define calibration, abstention, or mandatory human-review policies for low-confidence cases.*

[PENDING: centre-level variation in segmentation performance and its associations with scanner type, image quality, case mix and annotation protocol, and the calibration, abstention and mandatory human-review policies for low-confidence cases, provided by the image-analysis lead; the report-side review policy is described in the response to Reviewer 1, Comment 6.**]**

Reviewer 1, Comment 4

Data boundaries across tasks. *ZJH-8K is described as an external test set for segmentation and benign-malignant classification, while its malignant images are used for development of LNM and FTC/PTC tasks. This may be acceptable because the tasks differ, but the manuscript should provide a complete data-flow diagram showing which data were used for training, validation, threshold selection, and testing for each component.*

[PENDING: image-analysis components, to be merged into one data-flow figure and one response by the image-analysis lead.**]**

For the report-generation components, the data-flow panel (Supplementary Fig. S8) and Supplementary Table S12 show which data were used for development and which for evaluation. The SMU-HMC cohort (23,955 reports from 21,954 patients) consists of 20,400 single examinations, one per

patient, and a longitudinal follow-up set of 3,555 reports from 1,554 patients with 2–9 examinations each; the longitudinal set is released with the data and was not used for the template libraries or report-generation testing. Of the single examinations, the 400 test patients were held out at the patient level; the remaining 20,000 reports were used to develop the auxiliary tools, and 7,147 of them, selected by report completeness, to build the SMU-HMC template library. The KMVE template library was built from the training split (1,313 of its 1,719 reports, after excluding the 406 near-duplicates of test reports), its rules were checked on the validation split (246 cases), and the test split (492 cases) was used only for evaluation. ZJH-TS (150 reports) was used only for evaluation, with the SMU-HMC library, and TNVideo (145 videos with annotation-derived labels) only for the reader study. For the new ThyClinScore calibration, clinician ratings of SMU-HMC test reports are the development set for λ and η, which are locked before the KMVE and ZJH-TS ratings are analysed.

Reviewer 1, Comment 5

Validation of report quality and template dependence. *ThyClinScore is well motivated, but its validation primarily relies on correlation with an LLM judge. A stronger validation would include blinded assessment by multiple independent clinicians, inter-rater agreement, and sensitivity analyses for lesion-matching thresholds and score weights. The authors should also audit patient overlap, near-duplicate report text, and centre-specific template effects between template libraries and test sets.*

We thank the reviewer for these suggestions. We have added a blinded clinician assessment, sensitivity analyses for the lesion-matching threshold and all score weights, and an audit of the template libraries against the test sets.

Blinded clinician assessment. **[PENDING:** two clinicians independently rated 900 anonymised reports (100 cases per test set; the ThyroidXAgent report and two baseline reports per case) against the reference report on five dimensions (clinical accuracy, factual consistency, completeness, readability and clinical usability; 1-5 scale), blinded to the system; report inter-rater agreement (ICC and weighted kappa) and the Spearman correlation of ThyClinScore, BLEU, ROUGE-L and METEOR with the mean clinical-accuracy rating, with case-cluster bootstrap CIs.**]**

Sensitivity to the matching threshold and weights. We recomputed ThyClinScore for all models on the three test sets under 120 parameter configurations (Supplementary Table S13; KMVE results use the revised KMVE library described below). These varied the lesion-matching threshold (0.3–0.7), the weight of size relative to feature agreement in lesion matching (α_s 0.5–0.9), the size tolerances for gland and lesion measurements (τ 0.10–0.30 and 0.15–0.45), λ (0.1–0.5), η (0.3–0.7) and each of the six consistency weights and five completeness weights (0.5–1.5 times the default), singly and in three joint grids (λ with η, the threshold with α_s, and the two tolerances). ThyroidXAgent had the highest mean ThyClinScore in all 120 configurations in SMU-HMC, in 108 in KMVE and in 117 in ZJH-TS. In KMVE, LLaVA-Med, which reported no abnormality in any KMVE case, ranked first in the ten configurations with a matching threshold of 0.6 or 0.7 (differences −0.051 to −0.029; 95% CIs excluding zero), and GPT-5 ranked first in two of the five configurations with a threshold of 0.3 (differences −0.004 and −0.003; 95% CIs including zero). In ZJH-TS, ThyroidXAgent ranked second or third in three configurations, and all of these differences had 95% CIs that included zero. The ranking of all models changed little (median Kendall's τ_b 0.94 in SMU-HMC, 0.93 in KMVE and 0.94 in ZJH-TS), except in KMVE when the matching threshold was 0.3 or size and feature agreement were given equal or nearly equal weight in lesion matching (minimum τ_b 0.21).

Patient overlap. The SMU-HMC library was built after the 400 test patients had been excluded,

and none of its 7,147 source reports shares a report or patient identifier with the SMU-HMC test set. KMVE follows the official split, in which no training identifier matches a test identifier.

Near-duplicate report text. We compared each complete test reference report with every source report of its library, using the Jaccard similarity of character 3-grams after removing section headings, punctuation and spaces, and counted a test report as having a near-duplicate when the similarity reached 0.90 (Supplementary Tables S14 and S15). In SMU-HMC, 4 of the 400 test reports had a near-duplicate source report; these four reports and their five matching source reports all describe a normal thyroid without focal lesions in the standard wording used at the centre, and they differ only in the gland measurements. No ZJH-TS test report had a near-duplicate. In KMVE, 133 of the 492 test reports had a near-duplicate training report, and 105 had normalised text identical to that of a training report, most of them (83) describing a normal gland. We used the training and test split published with KMVE without modification. The frequent identical texts are probably because the released KMVE reports mask sizes and other measurements and retain only short descriptions of the findings, so that reports of normal glands and of simple nodules are often worded identically across patients. These identical texts led us to revise the KMVE library, as described next.

Revision of the KMVE library. We excluded the 406 training reports that were near-duplicates of KMVE test reports and rebuilt the KMVE library with the procedure used for the SMU-HMC library (our response to Reviewer 2, Comment 19), using the remaining 1,313 training reports to build the templates and the 246 validation reports to check them before the test set was scored. On the KMVE test set (Supplementary Table S16), ROUGE-L increased from 0.5880 to 0.6573 (difference 0.0694; 95% CI 0.0597 to 0.0791), BLEU-1 to BLEU-4 decreased by 0.035 to 0.068, METEOR decreased by 0.017, and ThyClinScore changed from 0.4407 to 0.4327 (−0.0080; 95% CI −0.0254 to +0.0098). ThyroidXAgent remained first among the methods of Supplementary Tables S8 and S9 on all six lexical metrics and on ThyClinScore. The KMVE results in the manuscript have been updated accordingly.

Removal of near-duplicate templates. From the SMU-HMC library, which is used for both SMU-HMC and ZJH-TS reports, we removed every template with a similarity of at least 0.90 to any sentence of at least 20 characters in the near-duplicate test reports identified above (5 templates) and regenerated all reports from the same tool outputs. None of these templates had been used in an SMU-HMC or ZJH-TS report, so these reports did not change. In a stricter analysis, we applied the same criterion to every test report and removed 166 templates. Lexical metrics were computed as in Supplementary Table S8 and ThyClinScore as in Supplementary Table S9 (Supplementary Table S17). ROUGE-L changed by −0.0016 (95% CI −0.0031 to −0.0004) in SMU-HMC and +0.0003 (−0.0015 to +0.0022) in ZJH-TS, and ThyClinScore by −0.0034 (−0.0067 to −0.0005) and +0.0043 (−0.0005 to +0.0107), respectively. In both analyses, ThyroidXAgent kept its rank among the methods of Supplementary Tables S8 and S9 on all six lexical metrics and on ThyClinScore in both test sets.

Centre-specific template effects. To assess the effect of the template wording, we exchanged the clause templates between the SMU-HMC library and the revised KMVE library while keeping the tool outputs and the report format of each test set (Supplementary Table S18 and Supplementary Fig. S9). KMVE reports were generated with the SMU-HMC library and, as in KMVE reports, kept only the gland, nodule and lymph-node sentences, with nodule sizes masked. SMU-HMC and ZJH-TS reports used gland and nodule sentences generated with the KMVE library and filled with the measured sizes, together with their original measurement sentence, lymph-node sentence and impression. The lexical metrics decreased substantially: ROUGE-L by 0.1247 (95% CI 0.1175 to 0.1318) in SMU-HMC, 0.2895 (0.2775 to 0.3017) in KMVE and 0.1045 (0.0936 to 0.1155) in ZJH-TS, and BLEU-4 by 0.119 to 0.276. ThyClinScore decreased much less: by 0.0484 (0.0409 to 0.0558) in SMU-HMC

and 0.0425 (0.0253 to 0.0592) in KMVE, and in ZJH-TS it changed by −0.0082 (−0.0217 to +0.0055). The remaining decrease reflects findings that the exchanged templates state differently or omit; for example, the KMVE templates, which follow the KMVE reports, do not describe the aspect ratio or internal echo of a nodule. Lexical metrics count shared word sequences, so they also fall when the same findings are worded or ordered differently. For example, one SMU-HMC nodule was described with the original templates as "hypoechoic area seen in the right thyroid lobe, about 43.5 × 13.0 mm, margin clear, shape regular, aspect ratio < 1, internal echo fairly homogeneous; CDFI: no flow signal within it" and with the KMVE templates as "a hypoechoic nodule is seen in the right lobe, about 43.5 × 13.0 mm, with a well-defined margin and a regular shape; CDFI shows no detectable flow signal" (translated from the Chinese reports). Both sentences state a hypoechoic nodule in the right lobe of 43.5 × 13.0 mm with a clear margin, a regular shape and no flow signal, but the ROUGE-L of the report fell from 0.4940 to 0.3783 and its BLEU-4 from 0.2172 to 0.1347, whereas its ThyClinScore was 0.5341 and 0.5349. ThyClinScore compares the findings after the report has been structured, which makes it much less sensitive to reporting style; this is why we designed it. All metrics in this audit were computed with one protocol for every condition.

Reviewer 1, Comment 6

Reader study and clinical safety. *The two-physician, 145-case crossover study supports initial workflow feasibility, but does not establish patient benefit. Results should be reported separately by physician, with case-order randomization, learning-effect analysis, error categories, and confidence intervals. More importantly, the system should define when it abstains or escalates to mandatory human review, and quantify the effect of these safety rules on missed diagnoses, false positives, and workload.*

[PENDING: reader-study results by physician with case-order randomisation, learning-effect analysis, error categories and confidence intervals, and the rules for abstention and mandatory human review, including the handling of missing gland measurements, with their effect on missed diagnoses, false positives and workload. These analyses await the reader-study data of the additional physicians and the clinician reference for the safety analysis.**]**

Response to Reviewer 2

Reviewer 2, Comment 1

Several key values are inconsistent across the abstract, main text, figures, and supplementary materials.

[PENDING: image-analysis values, to be merged into a single response to this comment by the image-analysis lead.**]**

We thank the reviewer and apologise for these inconsistencies. We checked every report-generation and reader-study value in the main text against the supplementary tables, the values shown in the figures and the original reader-study records (Supplementary Table S19), and made the following corrections.

Lexical metrics (Results 2.6). The values in the text now match Supplementary Table S8 and Fig. 5f: BLEU-1, BLEU-4 and ROUGE-L of 0.5924, 0.3381 and 0.5422 on SMU-HMC and of 0.5051, 0.2909 and 0.5480 on ZJH-TS. On KMVE, the values in the text (0.6209, 0.4465, 0.3596 and 0.5826 for BLEU-1, BLEU-4, METEOR and ROUGE-L) also differed from Supplementary Table S8; they are now those

of the revised KMVE library (0.5675, 0.4183, 0.3505 and 0.6573; our response to Reviewer 1, Comment 5), which also replace the earlier values in Supplementary Table S8 and Fig. 5f. We also state that ThyroidXAgent did not have the highest METEOR on SMU-HMC (0.3627 vs 0.3628 for Qwen3.5 Plus) or ZJH-TS (0.3293 vs 0.3508 for Qwen3.5 Plus).

ThyClinScore (Results 2.6). In the two sentences that report ThyroidXAgent's ThyClinScore, the values for SMU-HMC and ZJH-TS are corrected from 0.5238 and 0.4775 to 0.5189 and 0.4676, as in Supplementary Table S9 and Fig. 5f, and the KMVE value (0.4465) is replaced by that of the revised KMVE library (0.4327).

Reporting time (Results 2.6, Fig. 6d,e). The 27.4% reduction had been calculated over all 151 videos, whereas the other reader-study results use the 145 videos with annotation-derived labels. Over these 145 videos, the mean reporting time fell from 149.3 s to 110.5 s (2.5 to 1.8 min), a reduction of 26.0%; the text and Fig. 6d now give this value. The time savings of the two physicians differed (13.0% and 36.8%; Fig. 6e), and the text no longer describes them as similar.

Figure citation (Methods 4.5). The sentence on building template libraries from new report sources now cites Supplementary Fig. S6, which shows template-library construction, instead of Supplementary Fig. S4c, which shows video-input processing.

Reviewer 2, Comment 16

The ReAct-style planner–executor workflow lacks systematic failure-mode evaluation. The authors should report the frequency of invalid or unnecessary tool calls, hallucinated tasks, repeated calls, execution failures, retries, and rollbacks, and provide representative traces for successful cases and major failure modes, particularly cases with no nodules, multiple nodules, or conflicting intermediate evidence.

We thank the reviewer for this comment and agree that the original manuscript did not evaluate failure modes of the planner–executor workflow. We have therefore added a fully instrumented failure-mode audit (Supplementary Tables S20, S21, S22 and S23, and Supplementary Fig. S10).

Audit design. We audited 150 report-generation cases, 50 from each of SMU-HMC, KMVE and ZJH-TS. The cases were drawn from a sample stratified by the number of thyroid nodules or cysts stated in the reference report (0, 1, ≥2 or uncertain; Supplementary Table S20). Each case was run through the complete workflow three times, giving 450 runs and 2,347 tool calls. We used a root seed of 42 to deterministically derive a pre-assigned seed for each run. For every run we recorded the LLM requests and responses, each tool call with its arguments and raw return, the summary returned to the LLM, the case state before and after each call, and the final report. Each failure category was defined operationally (Supplementary Table S21) and assessed for all tool calls. Because each case was run three times, we also report how many cases were affected and whether an event recurred across a case's runs (Supplementary Table S22). Confidence intervals (CIs) were obtained by resampling cases within each dataset, keeping all three runs of each case.

Frequency of the requested failure categories (Supplementary Table S21).

- **Invalid calls.** Four of the 2,347 tool calls (0.17%; 95% CI 0.04–0.35%) were invalid. They occurred in four different cases, each in only one of its three runs. One call used a malformed tool name. The other three were gland-measurement requests that cited image identifiers that do not exist:

one was a zero-padded variant of an existing identifier, and two were numbered one beyond the last image. The executor did not process the non-existent identifiers.

- **Unnecessary calls.** We counted requests that listed images outside the tool's target region: 18 lymph-node screening calls that included non-neck images (0.77% of all calls; 95% CI 0.27–1.37%; 8 cases) and 66 gland-measurement calls that included neck images (2.81%; 1.96–3.70%; 32 cases). In both situations the executor passes only the eligible images to the tool, so the other images are simply not analysed in that call; nothing is deleted. For example, one screening request listed ten images, and only its five neck images were screened (Supplementary Fig. S10e). Because each of these calls still analysed eligible images, we regard them as imprecise requests rather than unnecessary calls. Region labels come from automated preprocessing.
- **Hallucinated tasks.** We defined a hallucinated task as a task that was planned or executed without support in the case's images. Every one of the 2,347 tool calls executed a task in the plan. We then checked each of the 2,348 planned tasks against the images available at planning time; for example, nodule segmentation requires an image flagged by preprocessing, and lymph-node screening requires a neck image. Three planned tasks (0.13%; 95% CI 0.00–0.29%) had no such support. All three were lymph-node screening planned for cases without neck images, each in a different case and in only one of its three runs. In two of these runs the task was never executed; in the third it was issued with an empty image list and returned nothing.
- **Repeated calls.** Across the 450 runs, apart from one retry triggered by a malformed tool name, no tool was called more than once within the same run.
- **Execution failures.** The only failed tool call was the malformed name. All 450 runs were completed. Three of them, in three different cases, were first interrupted by out-of-memory errors caused by an external factor (other jobs occupying GPU memory) and were re-run with the same inputs.
- **Retries and recovery.** The call with the malformed tool name was retried by the LLM at the next step, with the corrected name and the same arguments. In addition, the executor has a fallback for steps at which the LLM does not continue with the plan produced by the planner, that is, returns neither a tool call nor the completion signal: it then runs the next planned task that has not yet been performed. This fallback was triggered in two runs, each from a different case. In both, the LLM returned an empty response after gland measurement had completed; the executor ran the planned lymph-node screening, and the LLM then completed the run. Because the fallback can only run tasks listed in the plan, it cannot restore a step that the plan itself omits (see the omitted fusion step below).
- **Rollback.** The current system does not implement transactional rollback of tool actions.

Major failure modes and representative traces (Supplementary Table S23; Supplementary Fig. S10). Apart from the request errors above, all of which were contained, the main failure modes were ordering errors and omitted steps that the run-level success flags did not reveal:

- In 40 of the 320 fusion-tool calls (24 cases; in all three runs of 3 of them), fusion was executed before pixel spacing was available for the nodule-positive images, and spacing obtained later did not trigger re-fusion. In 4 SMU-HMC and ZJH-TS runs, this left 5 nodule descriptions without a size. The other 36 calls were in KMVE cases, whose reports use a mask in place of specific nodule sizes. In addition, in one run, fusion was executed before nodule feature classification.
- For the 43 cases in which preprocessing flagged no nodule, the workflow follows a plan that does not include spacing estimation. Consequently, gland measurement in these cases was performed without a pixel-spacing estimate.

- In one run, a segmented nodule was never fused because fusion was missing from the plan, and the report contained no description of that nodule.

The supplementary figure shows real traces for a case without nodules, a case with multiple nodules, conflicting intermediate evidence (a preprocessing-flagged image not confirmed by segmentation and excluded downstream, seen in all three runs of two cases), recovery from a malformed tool name, off-target images that were not processed, fusion before pixel spacing, and the omitted fusion step.

Reviewer 2, Comment 17

Agent workflow lacks concurrency control and conflict resolution mechanisms: When lymph-node screening and nodule analysis execute concurrently, how is duplication avoided if both detect the same hypoechoic structure in the lateral neck region?

We thank the reviewer for this question. In the report-generation workflow of ThyroidXAgent, lymph-node screening and nodule analysis do not run concurrently, and the workflow does not match objects between the two branches. Duplication is avoided by anatomical region: lymph-node screening analyses only neck-region images, and nodules assigned to the neck region are not described as thyroid nodules in the report. We have added this description to the Methods and quantified it in the 450 audit runs described in our response to Comment 16.

Execution order. Execution can be described at two levels, between tools and within a tool. Between tools, the only parallel step is preprocessing, in which three per-image classifiers (colour-Doppler detection, anatomical region classification and nodule-presence detection) run concurrently, each writing a separate field of the image record. Within a tool, the per-image tools, such as nodule segmentation, feature classification, pixel-spacing prediction and lymph-node screening, take a batch of images in one call and return one result per image. After preprocessing, the tools for different tasks, including nodule segmentation and lymph-node screening, are called one at a time by the executor. In the 450 audit runs, none of the 2,347 tool calls overlapped in time with another.

Separation by region. Preprocessing assigns each image to one region: a longitudinal or transverse view of the left or right lobe, a transverse view of both lobes or the isthmus, or the neck. By the definitions of the tools in the toolset, lymph-node screening accepts only neck images, and nodule segmentation analyses the images flagged by the nodule-presence detector, which can include neck images. The fusion tool groups nodules by lobe, so nodules in neck images are never merged with nodules in thyroid images. When the report is assembled, nodules in the neck region are not described as thyroid nodules.

What happened in the audit runs. The nodule-presence detector flagged 72 neck images in 27 of the 450 runs (6.0%; 95% CI 2.7–10.0%; 9 cases), and segmentation detected a structure in 69 of them. In 60 of these images (24 runs, 5.3%; 95% CI 2.0–9.3%; 8 cases), lymph-node screening was also positive, that is, both branches detected a structure in the same image. All 69 nodules segmented in neck images were fused as neck-region nodules, none was combined with a nodule from a thyroid image, and none appeared in a nodule description. They also did not change the TI-RADS category stated in the impression (24 runs; KMVE reports do not state a category).

The same separation held in the 1,042 test reports of the original study. There, 57 images in 35 reports were processed by both branches, and none of the 57 nodules fused from these images was described as a thyroid nodule.

Reviewer 2, Comment 18

The ThyClinScore weights λ and η directly affect report-quality assessment, but their selection, optimization dataset, and sensitivity analysis are not reported, raising concerns about overfitting. Although Figure 4c shows correlation with an LLM judge, validation against radiologist ratings and inter-rater reliability analysis are lacking.

We thank the reviewer. λ and η were set heuristically when the metric was designed ($\lambda = 0.3$, $\eta = 0.5$) and were not optimised on any dataset; the same applies to the other parameters, whose default values are listed in Supplementary Table S13. We now report how the results depend on these parameters and calibrate λ and η against clinician ratings.

Sensitivity analysis. λ was varied from 0.1 to 0.5 and η from 0.3 to 0.7, singly and in all 25 combinations, as part of 120 configurations that also varied the lesion-matching threshold, the size tolerances and all component weights (our response to Reviewer 1, Comment 5). ThyroidXAgent had the highest mean ThyClinScore in all 25 λ–η combinations in SMU-HMC and KMVE and in 24 of 25 in ZJH-TS; in the exception ($\lambda = 0.1$, $\eta = 0.7$), its differences from GPT-5 and Qwen3.5 Plus had 95% CIs that included zero. When λ or η was set to 0 or 1, ThyroidXAgent was not first in 3 of the 12 dataset–setting combinations. It ranked third in 2, both with $\eta = 1$: in ZJH-TS, its differences from GPT-5 and Qwen3.5 Plus had 95% CIs that included zero, and in KMVE, LLaVA-Med, which reported no abnormality in any KMVE case, ranked first (difference −0.039; 95% CI −0.058 to −0.020). It ranked second in KMVE with $\lambda = 1$, behind Qwen3.5 Plus (difference −0.0002; 95% CI −0.0122 to +0.0116).

Calibration and validation against clinician ratings. **[PENDING:** λ and η selected on the SMU-HMC clinician ratings by five-fold cross-validation over the 25 λ–η combinations, with all reports of a case in the same fold, and locked before the ZJH-TS and KMVE ratings were analysed; correlation with the ratings for the default and calibrated parameters on ZJH-TS and KMVE; inter-rater reliability (ICC and weighted kappa).**]**

Reviewer 2, Comment 19

The term "training-free" should be qualified. Although BM25 requires no parameter training, the workflow still depends on corpus selection, report decomposition, template construction, slot definitions, and clause-assembly rules. The authors should describe the extent of manual curation, template deduplication, slot-filling rules, and adaptation to centre-specific reporting styles. Clinician assessment of readability, factual consistency, and clinical usability is also needed beyond lexical-overlap metrics such as BLEU and ROUGE.

We thank the reviewer and agree. We no longer describe report generation as training-free. The report-realisation stage requires no additional model training or gradient-based fine-tuning, but it depends on template libraries and rules built from training reports. We now describe how these were built and used in the Methods and summarise them in Supplementary Table S24.

Corpus selection. For SMU-HMC, 20,400 reports were scored for completeness, and the 400 reports of the test patients were excluded before tokenisation. Of the remaining 20,000 reports, the 7,147 with a completeness score of at least 0.9 were used to build the library. The completeness score is the report-completeness component of ThyClinScore (Methods), computed on the structured version of each report: the weighted fraction of the required information on thyroid measurements, parenchyma, lesions, impression and lymph nodes that is present. The KMVE library was built from

the official KMVE training split (1,719 reports); in the revision, the 406 training reports that were near-duplicates of KMVE test reports were excluded, leaving 1,313 reports (our response to Reviewer 1, Comment 5). ZJH-TS reports were generated with the SMU-HMC library.

Decomposition and de-duplication. The SMU-HMC reports were divided into five clause categories (thyroid measurement, gland morphology, nodule findings, lymph-node findings and impression). In each clause, case-specific content, such as measurements, locations, echogenicity, vascularity and TI-RADS categories, was replaced by slot placeholders, so that each clause became a candidate template. Identical candidate templates were merged, and the number of clauses mapped to each template was recorded as its frequency. Infrequent templates were then removed: gland and measurement templates outside the 400 and 40 most frequent, which account for 81% and 99% of the gland and measurement clauses, respectively; lymph-node templates occurring fewer than 20 times; nodule templates occurring fewer than 3 times; and impression clauses occurring only once. The construction scripts also removed 29 nodule templates whose fixed echogenicity wording conflicted with the echogenicity slot and one measurement template with an incomplete value, and manual review removed 57 gland and measurement templates (see below). In this way, the 1,793 distinct gland templates were reduced to 358, and the 7,478 nodule clauses, which formed 4,656 distinct templates, were reduced to 204 nodule templates. The final SMU-HMC library has 4,774 entries: 400 general sentences (358 gland, 24 measurement and 18 lymph-node sentences), 204 nodule templates, 1,076 impression clauses and 3,094 whole impressions. The KMVE library was rebuilt in the same way during the revision (our response to Reviewer 1, Comment 5): the nodule sentences of these 1,313 training reports were converted into slot templates and merged with their frequencies, giving 82 nodule templates, and the 259 gland sentences were grouped by the gland size, parenchymal echo and parenchymal vascularity they state.

Manual curation. Manual review was applied to the 400 most frequent gland templates and the 40 most frequent measurement templates, and removed templates that are not typical of complete routine reports; all removed templates are listed in Supplementary Table S25. Of the gland templates, 42 were removed: 9 describing postoperative or residual glands and 33 that do not form a complete gland description (9 measurement sentences, 13 describing only glandular vascularity and 11 describing focal lesions). Of the measurement templates, 15 were removed: 7 with a note on an emergency or bedside examination, 4 from postoperative examinations, 3 with missing values and 1 noting that the measurement was unsatisfactory. All other steps were carried out by the construction scripts with the thresholds given above.

Slots and filling rules. The SMU-HMC templates contain slots for nodule location, size, composition, echogenicity, margin, shape, aspect ratio, internal echo, echogenic foci and vascularity, and for the TI-RADS category in the impression. A query is built from the structured tool outputs, the best-matching template is retrieved by term matching, and its slots are filled with the corresponding values. Nodules are described by lobe; when both lobes contain nodules, a bilateral template is tried first. The impression is assembled from impression clauses retrieved with the nodule sentences as queries, penalising clauses whose cystic content or calcification disagrees with the nodule sentence, and the TI-RADS category is then filled in. If no clause is retrieved, the most similar whole impression is used; if neither is available, a rule-based impression states the TI-RADS category of each lobe or that no focal lesion was seen. The lymph-node sentence is a fixed sentence reporting enlarged lymph nodes when screening is positive and is otherwise retrieved from the library. When the classifier provides no value, echogenicity is written as hypoechoic, the margin as smooth with a regular shape, and the aspect ratio as below 1. Vascularity is written as present when one of the

nodule's images is a colour-Doppler image and as absent otherwise. When no size is available, the size slot is removed. The KMVE templates contain slots for nodule location, number, echogenicity, margin, shape, echogenic foci and vascularity, which are filled with the same rules; nodule sizes are written as the masked size token of the KMVE reports, and the gland sentence is the most frequent KMVE sentence for the gland size, parenchymal echo and parenchymal vascularity stated by the gland-description tool. The handling of missing gland measurements is described in our response to Reviewer 1, Comment 6.

Centre-specific reporting styles. Each library reproduces the wording of its source reports, and libraries can be exchanged without changing the image-analysis tools or the evidence schema. SMU-HMC and ZJH-TS reports were generated with the SMU-HMC library, and KMVE reports with the KMVE library, which follows the findings-only format of that dataset. A library for a new centre is built from its reports with the same procedure and added as a separate profile, which is selected when a case is started (Supplementary Fig. S6). The dependence of the results on the template source is reported in our response to Reviewer 1, Comment 5.

Clinician assessment. **[PENDING:** results of the blinded five-dimension ratings (clinical accuracy, factual consistency, completeness, readability and clinical usability) by two clinicians of 900 reports; see the response to Reviewer 1, Comment 5.**]**

Reviewer 2, Comment 22

Please check the manuscript for grammatical errors, such as "ThyroidXAgent improved produced."

We thank the reviewer. The sentence now reads "ThyroidXAgent produced more clinically consistent reports". We also corrected "evidence-centerd" to "evidence-centred", the heading "Confilts" and the statement "The authors has no confilt of interests" to "Competing interests" and "The authors declare no competing interests", and the figure labels "Static Pipline" (Fig. 5g), "BlEU-4" (Fig. 4) and "Correlation with LLM a Judge score" (Fig. 4c), and "Lymph Node Metastatis" (Fig. 3b). **[AUTHOR:** state whether the full manuscript was language-edited, and by whom.**]**

Response to Reviewer 3

Reviewer 3, Major Comment 3

The definition of "28,458 non-overlapping test cases" is unclear. Please clearly describe the data hierarchy and confirm that all train/validation/test splits are patient disjoint. Besides, a simple flowchart showing the relationship of patient, examination, image/video, lesion and task would greatly improve the clarity.

[PENDING: definition of the 28,458 test cases, the data hierarchy, the patient-disjoint splits and the flowchart, provided by the image-analysis lead.**]**

For the report-generation components, the data used for development and for evaluation are shown in Supplementary Fig. S8 and Supplementary Table S12 (our response to Reviewer 1, Comment 4).

Reviewer 3, Major Comment 5

ThyClinScore is mainly validated by correlation with an LLM judge. This is not sufficient as a clinical gold standard. To strengthen the validity of the metric, please validate the metric against blinded ratings

from multiple radiologists and provide sensitivity analysis for the main weights and thresholds used. The manuscript should also acknowledge that ThyroidXAgent does not outperform all baselines across every semantic submetric.

We thank the reviewer and have addressed the three points.

Validation against blinded ratings. **[PENDING:** summary of the blinded clinician ratings, inter-rater agreement and correlation with ThyClinScore; details in the response to Reviewer 1, Comment 5.**]**

Sensitivity analysis. We recomputed ThyClinScore under 120 configurations of the weights and thresholds (our response to Reviewer 1, Comment 5, and Supplementary Table S13). ThyroidXAgent had the highest mean ThyClinScore in all 120 configurations in SMU-HMC, in 108 in KMVE and in 117 in ZJH-TS.

Submetrics. We now state in Results 2.6 that ThyroidXAgent does not lead on every semantic submetric. It had the highest ThyClinScore in all three test sets, but baselines led on the following submetrics (Supplementary Table S9). In SMU-HMC, LLaVA-Med had a lower FDR (0.1625 vs 0.2238). In KMVE (revised library), three baselines had a lower FDR (LLaVA-Med 0.0000, MedGemma 0.1877 and Gemini 2.5 Pro 0.3211 vs 0.3841); five had higher feature accuracy (GPT-5 0.6967, MedGemma 0.6820, GPT-4o 0.6608, Claude Sonnet 4.6 0.6605 and Qwen3.5 Plus 0.6154 vs 0.5320); Gemini 2.5 Pro had a higher lesion F1 (0.4648 vs 0.4071); five had higher completeness (Claude Sonnet 4.6 0.6707, MedGemma 0.6622, Qwen3.5 Plus 0.6525, LLaVA-Med 0.6508 and GPT-5 0.6506 vs 0.6326); and LLaVA-Med had higher consistency (0.5669 vs 0.5193). LLaVA-Med reported no abnormality in any KMVE case, which gives an FDR of 0. In ZJH-TS, LLaVA-Med had a lower FDR (0.2733 vs 0.2833), LLaVA-Med and the KMVE model had higher feature accuracy (0.6595 and 0.6288 vs 0.5820), and Qwen3.5 Plus and GPT-5 had higher consistency (0.4767 and 0.4735 vs 0.4564).

Reviewer 3, Minor Comment 1

ThyroidXAgent receives structured outputs from multiple specialized tools, whereas several multimodal LLM baselines appear to work more directly from images. The manuscript should clarify this difference and avoid interpreting these comparisons as a pure test of language-generation ability.

We thank the reviewer and agree. The methods in the report-generation comparison received different inputs, so the comparison is between complete systems rather than a test of language generation. We now state this in the Results and Methods and list the inputs of each method in Supplementary Table S26.

ThyroidXAgent. ThyroidXAgent analyses all images of a case (median 13 images per case in SMU-HMC, 11.5 in ZJH-TS and 2 in KMVE) with specialised tools for segmentation, feature classification, measurement and lymph-node screening. Its language model plans and calls these tools and receives their structured outputs, but it does not write the report. The report is assembled from the structured findings by retrieving templates from a library built from training reports: the SMU-HMC library for SMU-HMC and ZJH-TS, and the KMVE library for KMVE.

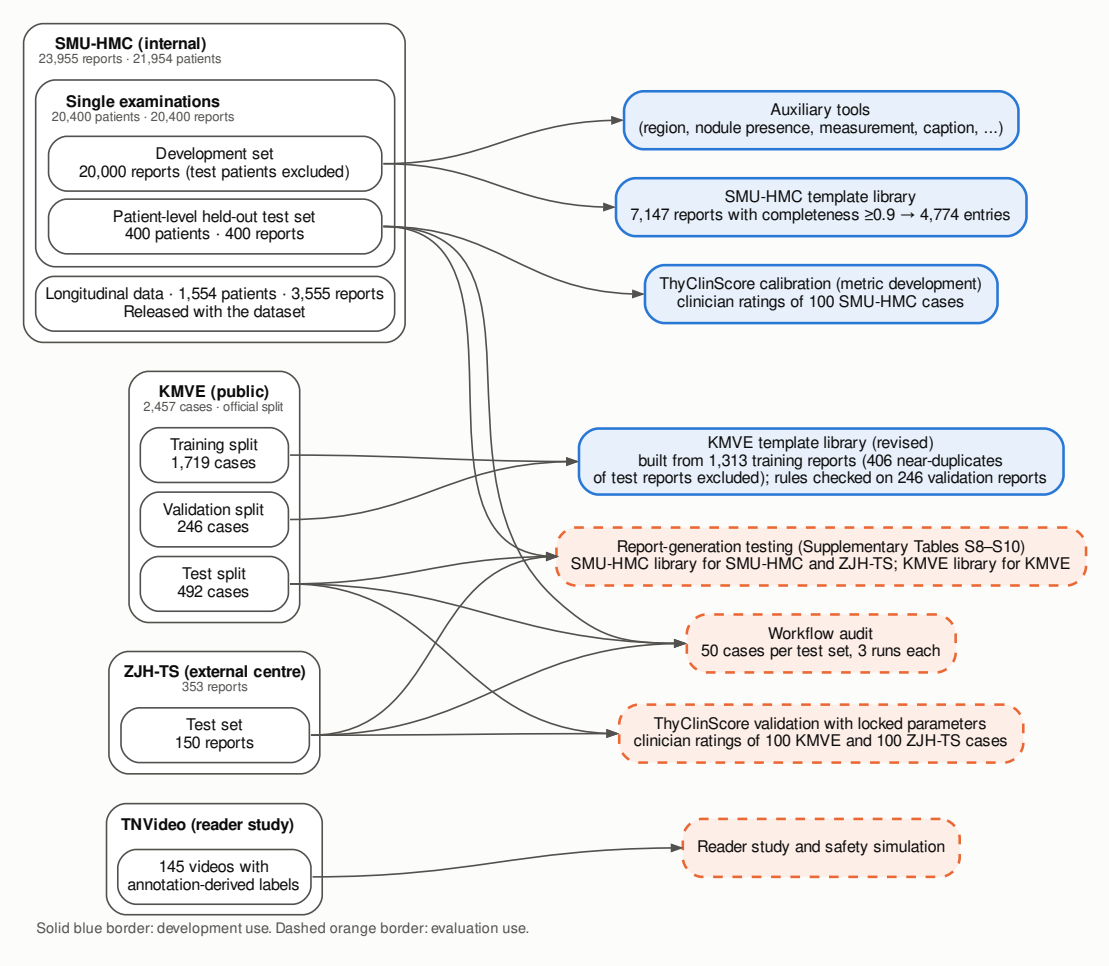
Baselines. The five general-purpose multimodal models (GPT-4o, GPT-5, Gemini 2.5 Pro, Qwen3.5 Plus and Claude Sonnet 4.6) and MedGemma received a prompt that specified the report structure and gave three example reports in the reporting format of the dataset (few-shot prompting). They also received up to five images sampled at random from each case in SMU-HMC and ZJH-TS, or the

two images of each case in KMVE. LLaVA-Med received two images and an English version of the prompt, and its output was translated into Chinese with GPT-4o-mini. The KMVE model, which was trained on the KMVE dataset to generate reports from images, received two images per case and was evaluated on SMU-HMC and ZJH-TS.

What the comparisons measure. The two comparisons address different questions. The comparison with the multimodal models and the KMVE model evaluates evidence-grounded report assembly as a whole: ThyroidXAgent derives structured evidence from all images with specialised tools and assembles the report from templates, whereas these models generate the report directly from a subset of the images (in KMVE, all methods received the same two images per case). Its result therefore reflects the benefit of grounding the report in tool-derived evidence, not the language-generation ability of the underlying model. The comparison with the static pipeline has matched inputs: the static pipeline has the same tools and templates as ThyroidXAgent, and the tools are called by fixed rules instead of being planned by the language model. Its result therefore reflects the contribution of adaptive tool routing (Fig. 5g and Supplementary Table S10).

Supplementary tables and figures cited in the responses

These tables and figures are added to the revised Supplementary Information; they are reproduced here in the order in which they are first cited.



Supplementary Fig. S8. Data flow of the report-generation components.

Left, the datasets and their partitions; right, their uses. Boxes with a solid blue border mark development uses and boxes with a dashed orange border evaluation uses. The SMU-HMC cohort comprises single examinations, one per patient, from which the 400 test patients were held out at the patient level, and a longitudinal set that is released with the data and was not used for the template libraries or report-generation testing (no arrows). The KMVE template library is the revised library described in our response to Reviewer 1, Comment 5. Details in Supplementary Table S12.

Supplementary Table S12. Data boundaries of the report-generation components.

Dataset	Partition	Used for	Not used for
SMU-HMC	Single examinations outside the test set (20,000 reports)	Auxiliary-tool development; SMU-HMC template library (7,147 reports with completeness $\geq$0.9)	Testing
SMU-HMC	Single examinations, held-out test set (400 patients, 400 reports)	Report-generation testing; ThyClinScore calibration on clinician ratings (100 cases); workflow audit (50 cases)	Template library
SMU-HMC	Longitudinal data (1,554 patients, 3,555 reports, 2–9 examinations per patient)	Released with the dataset	Template library; testing
KMVE	Training split (1,719 cases)	KMVE template library (1,313 reports, after excluding the 406 near-duplicates of test reports)	Testing
KMVE	Validation split (246 cases)	Checking the rules of the KMVE template library	Template library; testing
KMVE	Test split (492 cases)	Report-generation testing; ThyClinScore validation (100 cases); workflow audit (50 cases)	Template library
ZJH-TS	Test set (150 of 353 reports)	Report-generation testing with the SMU-HMC library; ThyClinScore validation (100 cases); workflow audit (50 cases)	Template library
TNVideo	145 videos with annotation-derived labels	Reader study and safety simulation	Template library; report-generation testing

Supplementary Table S13. Default values and ranges of the ThyClinScore parameters, and stability of the ThyroidXAgent ranking.

Panel A

Parameter	Default	Range analysed
λ (completeness vs consistency)	0.3	0.1–0.5
η (minimum lesion-detection gate)	0.5	0.3–0.7
Lesion-matching threshold	0.5	0.3–0.7
α_s (size weight in lesion matching; $\beta_f = 1 - \alpha_s$)	0.7	0.5–0.9
τ, gland size tolerance	0.2	0.1–0.3
τ, lesion size tolerance	0.3	0.15–0.45
Six consistency weights (thyroid size, lesion size, vascularity, lesion detection, lesion features, morphology)	0.2 / 0.15 / 0.15 / 0.15 / 0.2 / 0.15	0.5–1.5 $\times$ default
Five completeness weights (thyroid measurements, parenchyma, lesions, impression, lymph nodes)	0.25 / 0.25 / 0.35 / 0.1 / 0.05	0.5–1.5 $\times$ default

Panel B

Test set	Configurations with ThyroidXAgent highest	ThyroidXAgent rank	Median Kendall τ_b (minimum)	Configurations without ThyroidXAgent first
SMU-HMC	120/120	1	0.94 (0.89)	None
KMVE	108/120	1–3	0.93 (0.21)	12 configurations; LLaVA-Med first in 10 (difference −0.051 to −0.029; 95% CIs exclude 0); GPT-5 first in 2 (difference −0.004 to −0.003; 95% CIs include 0)
ZJH-TS	117/120	1–3	0.94 (0.83)	3 configurations; GPT-5 first in 2 (difference −0.003 to −0.001; 95% CIs include 0); Qwen3.5 Plus first in 1 (difference −0.001; 95% CIs include 0)

Panel A lists the default values, set heuristically and not optimised on any dataset. Panel B covers the 120 main configurations (single-parameter changes, three 5×5 joint grids and equal-weight variants); 4 boundary settings (λ or η equal to 0 or 1) are reported in the response to Reviewer 2, Comment 18. Differences and 95% CIs: paired percentile bootstrap (2,000 replicates, seed 20261011), resampling ZJH-TS reports by case and SMU-HMC and KMVE reports individually. GPT-5 on ZJH-TS uses the recomputed structured extraction (mean ThyClinScore 0.4514 at the defaults, vs 0.4472 in Supplementary Table S9).

Supplementary Table S14. Overlap audit between template libraries and test sets.

Test set (library)	Identifier overlap with library sources	Test reports with a near-duplicate source report (Jaccard ≥0.90)	Of which identical normalised text	Source reports involved	Removed: near-duplicate reports / all test reports	ROUGE-L change after exchanging clause templates
SMU-HMC (SMU-HMC library)	Report and patient identifiers: 0/400	4/400	0	5	5 / 166 templates (SMU-HMC library)	−0.1247 (−0.1318 to −0.1175)
ZJH-TS (SMU-HMC library)	Not applicable (no ZJH-TS report used to build a library)	0/150	0	0	5 / 166 templates (SMU-HMC library)	−0.1045 (−0.1155 to −0.0936)
KMVE (KMVE library)	Training/test identifiers: 0/492	133/492	105	406	406 training reports, excluded before the revised KMVE library was built	−0.2895 (−0.3017 to −0.2775)

Near-duplicate: Jaccard similarity of character 3-grams between each complete test reference report and each complete source report of its library, after removing section headings, punctuation and spaces; a test report counts when its highest similarity is ≥0.90 (details in Supplementary Table S15). Removal: from the SMU-HMC library, templates with similarity ≥0.90 to any sentence of at least 20 characters in the near-duplicate test reports or, in the stricter analysis, in any test report (results in Supplementary Table S17); for KMVE, the training reports that were near-duplicates of test reports were excluded before the revised KMVE library was built (Supplementary Table S16). Exchange: clause templates exchanged between the SMU-HMC library and the revised KMVE library, keeping the tool outputs and report format of each test set (Supplementary Table S18); ROUGE-L difference with 95% CI from a paired bootstrap over reports (10,000 replicates, seed 42).

Supplementary Table S15. Near-duplicates between test reports and template source reports (complete reports, character 3-gram Jaccard ≥0.90).

Panel A

SMU-HMC test report	Jaccard	Standard wording	Measurements in the test report	Measurements in the source report
Test report 1	0.911	A	left lobe 34×12×13 mm, right lobe 39×11×13 mm, isthmus 2 mm	left lobe 39×11×13 mm, right lobe 39×12×14 mm, isthmus 2 mm
Test report 2	0.910	A	left lobe 41×11×14 mm, right lobe 43×11×14 mm, isthmus 3 mm	left lobe 41×12×14 mm, right lobe 43×11×15 mm, isthmus 3 mm
Test report 3	0.918	B	left lobe 50×13×16 mm, right lobe 56×12×14 mm, isthmus 2 mm	left lobe 52×12×14 mm, right lobe 50×13×16 mm, isthmus 2 mm
Test report 4	0.910	B	left lobe 47×12×15 mm, right lobe 48×13×16 mm, isthmus 2 mm	left lobe 47×12×14 mm, right lobe 48×13×16 mm, isthmus 2 mm
Test report 4	0.910	B	left lobe 47×12×15 mm, right lobe 48×13×16 mm, isthmus 2 mm	left lobe 45×12×15 mm, right lobe 47×13×16 mm, isthmus 2 mm

Panel B

KMVE test reports	Normal gland	Nodule description	Postoperative	Total
Identical normalised text	83	15	7	105
Similarity ≥0.90, not identical	3	25	0	28
Total	86	40	7	133

Panel A lists all SMU-HMC near-duplicates (4 test reports, one of which matched two source reports); no ZJH-TS test report had a near-duplicate. In each pair the text apart from the gland measurements is identical and is one of two standard wordings for a normal examination: A, “甲状腺形态大小正常，腺包膜光滑，腺实质回声均匀，未见明确光团及液性暗区，CDFI：内见少量纤细分枝状彩色血流信号。双侧颈动脉周围未见肿大淋巴结回声。甲状腺未见局灶性病变。” (Normal thyroid shape and size, smooth capsule, homogeneous parenchymal echo, no definite mass or fluid area; CDFI: a small amount of fine branching colour flow signal. No enlarged lymph nodes around the bilateral carotid arteries. No focal thyroid lesion.); B, “甲状腺形态大小正常，腺包膜光滑，腺实质回声均匀，未见明确光团及液性暗区，内见少量纤细分枝状彩色血流信号。双侧颈动脉周围未见异常肿大淋巴结。甲状腺未见局灶性病变。” (Normal thyroid shape and size, smooth capsule, homogeneous parenchymal echo, no definite mass or fluid area, with a small amount of fine branching colour flow signal. No abnormally enlarged lymph nodes around the bilateral carotid arteries. No focal thyroid lesion.). Panel B classifies the 133 KMVE test reports with a near-duplicate among the 406 matching training reports; 68 of the identical texts are the same standard description of a normal thyroid. Sizes are masked in the released KMVE reports.

Supplementary Table S16. Lexical metrics and ThyClinScore of ThyroidXAgent on the KMVE test set before and after revision of the KMVE library, and its rank among the methods of Supplementary Tables S8 and S9.

Metric	Original KMVE generator (submitted manuscript)	Revised library	Difference (95% CI)	Rank: original / revised	Best baseline (Table S8/S9)
BLEU-1	0.6357	0.5675	−0.0682	1 / 1	Qwen3.5 Plus 0.5425
BLEU-2	0.5606	0.5124	−0.0482	1 / 1	Qwen3.5 Plus 0.4243
BLEU-3	0.5008	0.4619	−0.0389	1 / 1	Qwen3.5 Plus 0.3415
BLEU-4	0.4535	0.4183	−0.0352	1 / 1	Qwen3.5 Plus 0.2783
METEOR	0.3672	0.3505	−0.0167	1 / 1	Claude Sonnet 4.6 0.3284
ROUGE-L	0.5880	0.6573	+0.0694 (+0.0597 to +0.0791)	1 / 1	Qwen3.5 Plus 0.5066
ThyClinScore	0.4407	0.4327	−0.0080 (−0.0254 to +0.0098)	1 / 1	MedGemma 0.3932
FDR (lower is better)	0.4858	0.3841	−0.1016 (−0.1402 to −0.0640)	6 / 4	LLaVA-Med 0.0000
Feature accuracy	0.7366	0.5320	−0.1919 (−0.2587 to −0.1231)	1 / 6	GPT-5 0.6967
Lesion F1	0.3623	0.4071	+0.0449 (+0.0138 to +0.0764)	4 / 2	Gemini 2.5 Pro 0.4648
Completeness	0.6416	0.6326	−0.0091 (−0.0130 to −0.0051)	6 / 6	Claude Sonnet 4.6 0.6707
Consistency	0.5654	0.5193	−0.0461 (−0.0687 to −0.0230)	2 / 2	LLaVA-Med 0.5669

Original generator: whole training sentences retrieved and used without filling in the tool outputs. Revised library: built with the SMU-HMC procedure from the 1,313 KMVE training reports that remained after the 406 near-duplicates of test reports (Supplementary Table S15) had been excluded; training sentences converted into slot templates (82 nodule templates; gland sentences grouped by gland size, parenchymal echo and parenchymal vascularity), slots filled from the tool outputs, nodule sizes masked as in KMVE reports, and details not provided by the tools (such as the position of a nodule within the lobe) not written. The rules were checked on the 246 validation reports (reports regenerated from the facts stated in them: ROUGE-L 0.7092) before the test set was scored. Lexical metrics with the scoring script of Supplementary Table S8 and ThyClinScore as in Supplementary Table S9; the original values reproduce those tables exactly. Differences: paired bootstrap over reports (10,000 replicates, seed 42); none is given for BLEU and METEOR, which are corpus-level scores. The revised reports contain 21,884 tokens, 75% of the reference reports (29,291), compared with 28,463 for the original reports (BLEU brevity penalty 0.713 vs 0.971). The margin of the first nodule agreed with the reference report in 156 of 290 reports (253 for the original reports, which never stated an unclear margin), and the parenchymal echo in 174 of 474 (297).

Supplementary Table S17. Lexical metrics and ThyClinScore of ThyroidXAgent before and after removing near-duplicate templates from the SMU-HMC library, and its rank among the methods of Supplementary Tables S8 and S9.

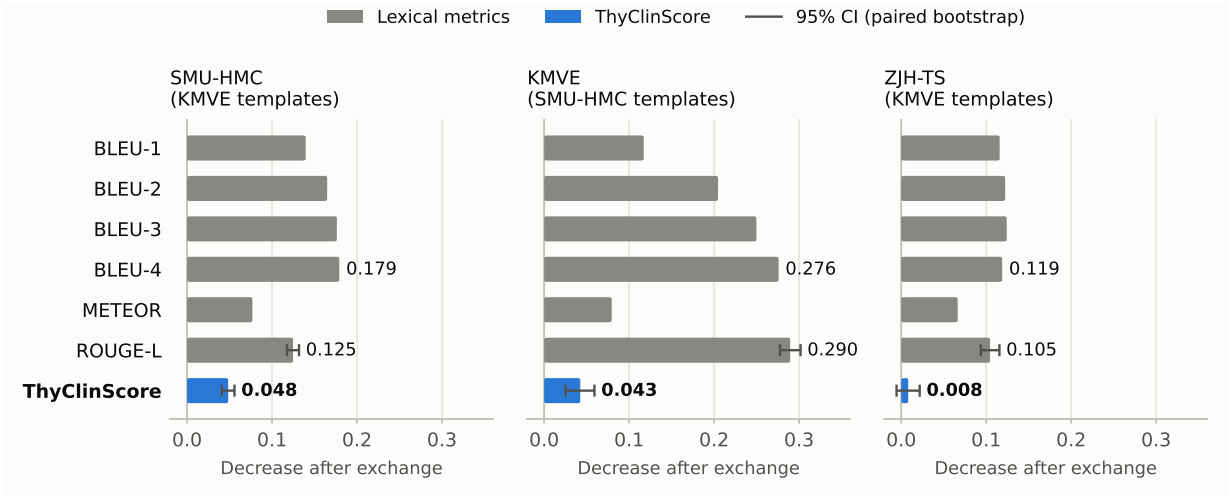
Test set	Metric	Before removal	After removal for near-duplicate reports	After removal for all test reports	Rank: before / near-duplicate removal / all-report removal	Best baseline (Table S8/S9)
SMU-HMC	BLEU-1	0.5924	0.5924	0.5907	1 / 1 / 1	Qwen3.5 Plus 0.4483
SMU-HMC	BLEU-2	0.4806	0.4806	0.4781	1 / 1 / 1	Qwen3.5 Plus 0.3623
SMU-HMC	BLEU-3	0.4006	0.4006	0.3968	1 / 1 / 1	Qwen3.5 Plus 0.2959
SMU-HMC	BLEU-4	0.3381	0.3381	0.3330	1 / 1 / 1	Qwen3.5 Plus 0.2427
SMU-HMC	METEOR	0.3627	0.3627	0.3611	2 / 2 / 2	Qwen3.5 Plus 0.3628
SMU-HMC	ROUGE-L	0.5422	0.5422	0.5406	1 / 1 / 1	Qwen3.5 Plus 0.5147
SMU-HMC	ThyClinScore	0.5189	0.5189	0.5154	1 / 1 / 1	Qwen3.5 Plus 0.4883
ZJH-TS	BLEU-1	0.5051	0.5051	0.5060	1 / 1 / 1	Qwen3.5 Plus 0.4964
ZJH-TS	BLEU-2	0.4137	0.4137	0.4146	1 / 1 / 1	Qwen3.5 Plus 0.3965
ZJH-TS	BLEU-3	0.3447	0.3447	0.3455	1 / 1 / 1	Qwen3.5 Plus 0.3196
ZJH-TS	BLEU-4	0.2909	0.2909	0.2915	1 / 1 / 1	Qwen3.5 Plus 0.2581
ZJH-TS	METEOR	0.3293	0.3293	0.3297	4 / 4 / 4	Qwen3.5 Plus 0.3508
ZJH-TS	ROUGE-L	0.5480	0.5480	0.5483	1 / 1 / 1	Qwen3.5 Plus 0.5314
ZJH-TS	ThyClinScore	0.4676	0.4676	0.4719	1 / 1 / 1	Qwen3.5 Plus 0.4520

SMU-HMC library (used for SMU-HMC and ZJH-TS): templates with similarity $\geq$0.90 to any sentence of at least 20 characters in the near-duplicate test reports of Supplementary Table S15 (5 templates) or, in the stricter analysis, in any test report (166 templates) were removed. Lexical metrics were computed with the scoring script of Supplementary Table S8 and ThyClinScore as in Supplementary Table S9 (same scoring script and structuring prompt, GPT-4o-mini, starting from the structured reports of the selected run so that only changed reports were structured again). Ranks are among all methods of Table S8 (lexical metrics) or Table S9 (ThyClinScore). Differences in the response: paired bootstrap over reports (10,000 replicates, seed 42).

Supplementary Table S18. Change in the lexical metrics and ThyClinScore of ThyroidX-Agent after exchanging clause templates between the SMU-HMC library and the revised KMVE library.

Test set (templates used)	Metric	Original templates	Exchanged templates	Change (95% CI)
SMU-HMC (revised KMVE templates)	BLEU-1	0.5924	0.4528	−0.1396
SMU-HMC (revised KMVE templates)	BLEU-2	0.4806	0.3157	−0.1649
SMU-HMC (revised KMVE templates)	BLEU-3	0.4006	0.2243	−0.1763
SMU-HMC (revised KMVE templates)	BLEU-4	0.3381	0.1589	−0.1792
SMU-HMC (revised KMVE templates)	METEOR	0.3627	0.2858	−0.0769
SMU-HMC (revised KMVE templates)	ROUGE-L	0.5422	0.4175	−0.1247 (−0.1318 to −0.1175)
SMU-HMC (revised KMVE templates)	ThyClinScore	0.5189	0.4705	−0.0484 (−0.0558 to −0.0409)
KMVE (SMU-HMC templates)	BLEU-1	0.5675	0.4503	−0.1172
KMVE (SMU-HMC templates)	BLEU-2	0.5124	0.3078	−0.2046
KMVE (SMU-HMC templates)	BLEU-3	0.4619	0.2121	−0.2498
KMVE (SMU-HMC templates)	BLEU-4	0.4183	0.1424	−0.2759
KMVE (SMU-HMC templates)	METEOR	0.3505	0.2710	−0.0795
KMVE (SMU-HMC templates)	ROUGE-L	0.6573	0.3678	−0.2895 (−0.3017 to −0.2775)
KMVE (SMU-HMC templates)	ThyClinScore	0.4327	0.3903	−0.0425 (−0.0592 to −0.0253)
ZJH-TS (revised KMVE templates)	BLEU-1	0.5051	0.3893	−0.1158
ZJH-TS (revised KMVE templates)	BLEU-2	0.4137	0.2914	−0.1223
ZJH-TS (revised KMVE templates)	BLEU-3	0.3447	0.2206	−0.1241
ZJH-TS (revised KMVE templates)	BLEU-4	0.2909	0.1721	−0.1188
ZJH-TS (revised KMVE templates)	METEOR	0.3293	0.2629	−0.0664
ZJH-TS (revised KMVE templates)	ROUGE-L	0.5480	0.4435	−0.1045 (−0.1155 to −0.0936)
ZJH-TS (revised KMVE templates)	ThyClinScore	0.4676	0.4594	−0.0082 (−0.0217 to +0.0055)

KMVE: reports generated with the SMU-HMC library, keeping only the gland, nodule and lymph-node sentences and masking nodule sizes, as in KMVE reports; original values are those of the revised KMVE library. SMU-HMC and ZJH-TS: gland and nodule sentences generated with the revised KMVE library and filled with the measured sizes, with the original measurement sentence, lymph-node sentence and impression. In 21 of these 550 reports, 36 sizes remained unfilled because the tool output gave no size for the nodule; the reports were scored as generated. All reports use the same tool outputs as Supplementary Tables S8 and S9. Lexical metrics and ThyClinScore were computed as in Supplementary Table S17. 95% CIs of the change: paired bootstrap over reports (10,000 replicates, seed 42), given for ROUGE-L and ThyClinScore; BLEU and METEOR are corpus-level scores. The changes are plotted in Supplementary Fig. S9.



Supplementary Fig. S9. Change in lexical metrics and ThyClinScore after exchanging clause templates between the SMU-HMC library and the revised KMVE library.

Bars show the decrease of each metric after the exchange, that is, its value with the original templates minus its value with the exchanged templates (SMU-HMC templates for SMU-HMC and ZJH-TS; revised KMVE templates for KMVE); error bars show 95% CIs for ROUGE-L and ThyClinScore from a paired bootstrap over reports (10,000 replicates, seed 42). BLEU and METEOR are corpus-level scores without CIs. Tool outputs and report format as in Supplementary Tables S8 and S9; values in Supplementary Table S18.

Supplementary Table S19. Corrected report-generation and reader-study values.

Location	Original	Corrected	Source of the corrected value
Results 2.6, SMU-HMC BLEU-1/BLEU-4/ROUGE-L	0.5961 / 0.3405 / 0.5450	0.5924 / 0.3381 / 0.5422	Supplementary Table S8; Fig. 5f
Results 2.6, KMVE BLEU-1/BLEU-4/METEOR/ROUGE-L	0.6209 / 0.4465 / 0.3596 / 0.5826 (Supplementary Table S8: 0.6357 / 0.4535 / 0.3672 / 0.5880)	0.5675 / 0.4183 / 0.3505 / 0.6573 (revised KMVE library)	Supplementary Table S8 with the revised KMVE library (response to Reviewer 1, Comment 5); Fig. 5f
Results 2.6, ZJH-TS BLEU-1/BLEU-4/ROUGE-L	0.5134 / 0.2942 / 0.5529	0.5051 / 0.2909 / 0.5480	Supplementary Table S8; Fig. 5f
Results 2.6, ThyClinScore of ThyroidXAgent (two sentences; SMU-HMC/KMVE/ZJH-TS)	0.5238 / 0.4465 / 0.4775 (Supplementary Table S9: 0.5189 / 0.4407 / 0.4676)	0.5189 / 0.4327 / 0.4676 (KMVE: revised library)	Supplementary Table S9 with the revised KMVE library; Fig. 5f
Results 2.6 and Fig. 6d, reduction in reporting time	27.4% (151 videos)	26.0% (149.3 s to 110.5 s; 145 labelled videos)	Original reader-study records
Results 2.6, time savings of the two physicians	similar	13.0% and 36.8%	Original reader-study records; Fig. 6e
Methods 4.5, figure reference for template-library construction	Supplementary Fig. S4c (video-input processing)	Supplementary Fig. S6 (template-library construction)	Supplementary Fig. S4 and S6 captions

Supplementary Table S20. Composition of the 150 audited cases by reference-report lesion count.

Dataset	0	1	2+	U (count uncertain)	Total
SMU-HMC	11	12	23	4	50
KMVE	12	11	23	4	50
ZJH-TS	1	11	38	0	50

Strata are the number of thyroid nodules or cysts stated in the existing reference report (cervical lymph nodes excluded). U, count could not be determined from the report text. Before execution, three cases whose inputs exceeded the runtime input-length limit were replaced by cases from the same dataset and stratum.

Supplementary Table S21. Failure-mode audit (150 cases × 3 runs = 450 runs; 2,347 tool calls).

Category	Operational definition	n/N	Frequency (95% CI)	Interpretation
1. Invalid calls (any)	Unregistered tool name, or an image ID absent from the case record	4/2347 calls; 4/450 runs	0.17% (0.04–0.35%) of calls; 0.89% (0.22–1.78%) of runs	One malformed tool name (corrected at the next call) and three requests with non-existent IDs (one zero-padded variant of an existing ID; two one beyond the last ID); non-existent IDs were not processed
2. Unnecessary calls (technical proxy)	Request listing images outside the tool's target region; the executor processes only eligible images	lymph-node screening with non-neck images 18/2347 calls (44 references); gland measurement with neck images 66/2347 calls (211 references)	0.77% (0.27–1.37%); 2.81% (1.96–3.70%)	Each of these calls also had eligible targets that were processed, so the call had a valid purpose
3. Hallucinated tasks	Task planned or executed without support in the case's images (e.g. segmentation without a preprocessing-flagged image; lymph-node screening without a neck image)	executed calls without a plan task 0/2347; planned tasks without image support 3/2348 (3 cases, one run each)	0.13% (0.00–0.29%)	All three were lymph-node screening planned for cases without neck images; two were never executed and one was issued with an empty image list and returned nothing
4. Repeated calls	Same tool with identical arguments (exact); or same tool, same targets, unchanged inputs and earlier results still present (semantic)	exact 0/2347; semantic 0/2347; tool invoked twice in a run 1/450 runs	0 (no interval for zero counts)	The only repeated tool was the retry after the malformed name

(continued)

Category	Operational definition	n/N	Frequency (95% CI)	Interpretation
5. Execution failures	Tool returned an error (call level); no completion signal (run level, first attempts)	1/2347 calls; first attempts 447/450 completed	99.3% (98.4–100.0%) completion	The call-level failure is the malformed name. The 3 incomplete first attempts were interrupted when other jobs occupied GPU memory (confirmed by the run operator)
6. Retries and recovery	Counted separately by mechanism	name correction 1; plan-guided supplementary calls 2; offline re-executions 3	—	Plan-guided calls followed an LLM turn with neither a tool call nor a completion signal and only cover planned tasks
7. Rollback	Transactional undo of tool effects	not implemented	—	—

CIs: percentile bootstrap resampling cases within each dataset and keeping all three runs and calls of each case (10,000 replicates, seed 42); none reported for zero or single-event counts.

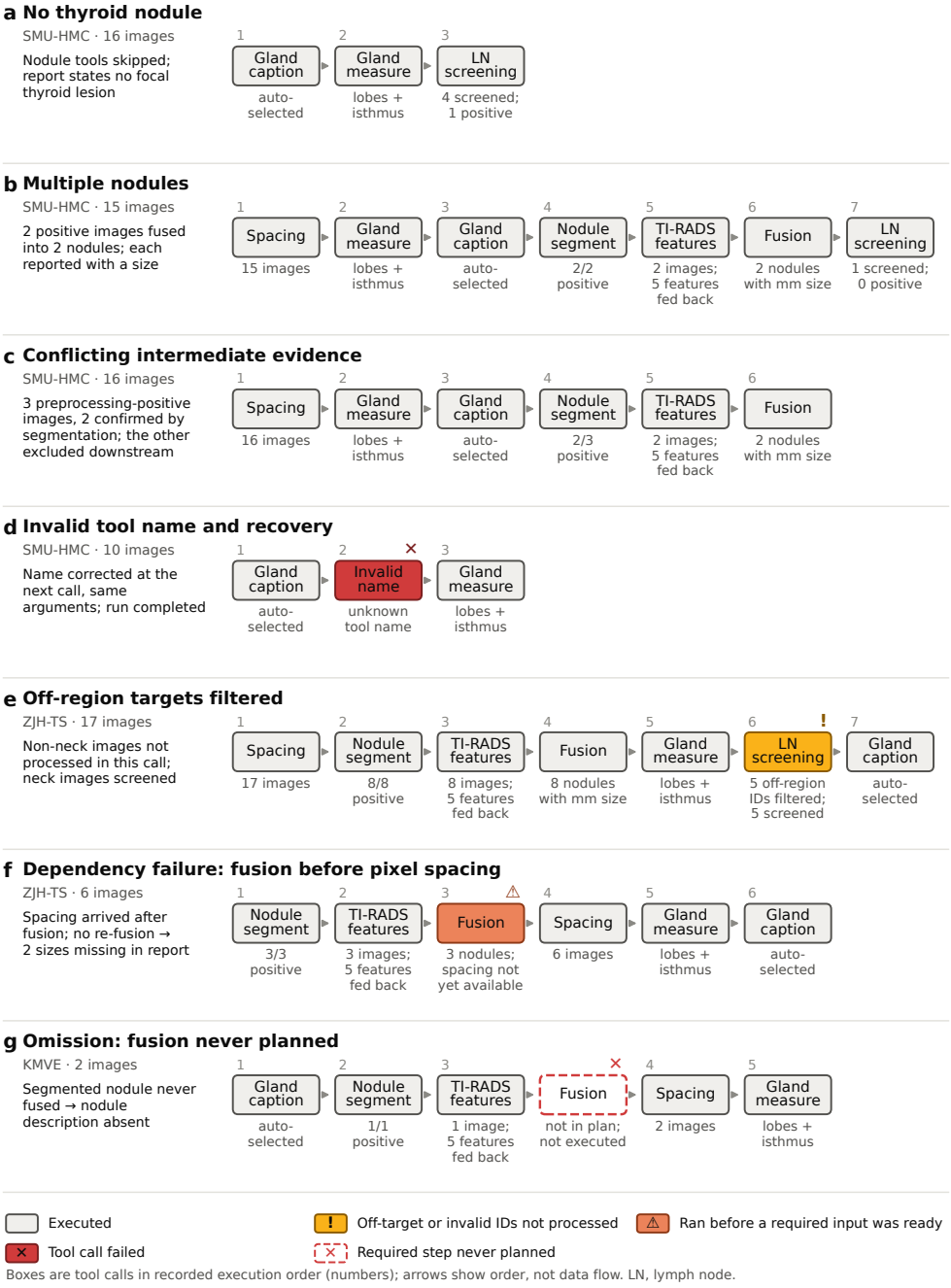
Supplementary Table S22. Distribution of each event across cases and their three runs (150 cases × 3 runs).

Event	Events	Runs affected	Cases affected	Cases with the event in 1 / 2 / all 3 runs	Datasets (cases)
Invalid call (unregistered name or non-existent image ID)	4 calls	4	4	4 / 0 / 0	SMU-HMC 4
Lymph-node screening request with non-neck images	18 calls	18	8	2 / 2 / 4	SMU-HMC 2; ZJH-TS 6
Gland-measurement request with neck images	66 calls	66	32	9 / 12 / 11	KMVE 1; SMU-HMC 17; ZJH-TS 14
Request with no processable target	1 call	1	1	1 / 0 / 0	SMU-HMC 1
Plan-guided supplementary call	2 calls	2	2	2 / 0 / 0	SMU-HMC 2
Fusion before pixel spacing	40 calls	40	24	11 / 10 / 3	KMVE 20; SMU-HMC 2; ZJH-TS 2
Fusion before feature classification	1 call	1	1	1 / 0 / 0	ZJH-TS 1
Gland measurement without any spacing estimate	129 runs	129	43	0 / 0 / 43	KMVE 18; SMU-HMC 24; ZJH-TS 1
Segmented nodule never fused	1 run	1	1	1 / 0 / 0	KMVE 1
Preprocessing-flagged image not confirmed by segmentation	6 runs	6	2	0 / 0 / 2	SMU-HMC 1; ZJH-TS 1
Planned task without image support (hallucinated task)	3 planned tasks	3	3	3 / 0 / 0	KMVE 1; SMU-HMC 1; ZJH-TS 1
First attempt interrupted (external GPU-memory contention)	3 first attempts	3	3	3 / 0 / 0	SMU-HMC 2; ZJH-TS 1

An event present in all three runs of a case recurs with that case's inputs; an event in one run only is sporadic. Each run of a case used a different pre-assigned seed.

Supplementary Table S23. Dependency, ordering and omission events not reflected in success flags.

Observation	n/N	Visible consequence
Fusion run before pixel spacing for all positive images	40/320 fusion calls	4 SMU-HMC/ZJH-TS runs with 5 nodule descriptions lacking a size; the other 36 calls (KMVE) produced no comparable empty-size sentence because KMVE reports use the dataset's size mask
Fusion run before feature classification	1/320 fusion calls	Classifications were not yet available when fusion ran, and later classification did not trigger re-fusion
Fusion inputs incomplete (either of the above)	41/320	12.8% (8.8–17.1%)
Gland measurement without any pixel-spacing estimate	129/450 runs (43 cases)	28.7% (22.0–35.3%); all runs on the plan used for cases without preprocessing-flagged nodules, which does not schedule spacing estimation
Segmented nodule never fused (step omitted from plan)	1/450 runs	Nodule description absent from the report
Preprocessing-positive image not confirmed by segmentation	6/450 runs (2 cases)	Image excluded from classification and fusion



Supplementary Fig. S10. Representative execution traces of the planner–executor workflow.

Each row shows one run from the audit (150 cases × 3 runs). Boxes are tool calls in the recorded execution order, given by the small numbers; arrows indicate order only, not data flow. The text under each box summarises the recorded arguments and returns, and the left column gives the dataset, the number of input images and the observed outcome. Fill colour and symbol together encode the execution status (legend).

a, No thyroid nodule. No image was flagged by preprocessing, so segmentation, feature classification and fusion were not scheduled. Lymph-node screening of the four neck images returned one positive image, which the report reflects.
b, Multiple nodules. Two preprocessing-flagged images were confirmed by segmentation, classified, and fused into two nodules, each reported with a size.
c, Conflicting intermediate evidence. Of three preprocessing-flagged images, two were confirmed by segmentation; the third was not passed to classification or fusion. The same disagreement occurred in all three runs of this case and of one other case.

d, Invalid tool name and recovery. The LLM emitted a malformed tool name and the executor returned an unknown-tool error; the next call used the registered name with identical arguments.
e, Off-region targets not processed. A lymph-node screening request listed five images whose preprocessing region was not the neck. The executor passed only the neck images to the screening tool; the other five were not processed in this call.
f, Dependency failure. Fusion ran before pixel-spacing estimates were available, so the fused nodules had no millimetre sizes. Spacing was computed later, but fusion was not repeated, and the report contains two nodule descriptions without sizes.
g, Omission. A nodule was confirmed by segmentation and classified, but fusion was neither planned nor executed (dashed box), and the report contains no nodule description. Plan-guided recovery could not add the step because it was absent from the plan.

Frequencies of each failure mode are given in Supplementary Tables S21 and S23. Case identifiers, file paths and report text are omitted. LN, lymph node.

Supplementary Table S24. Sources, composition and construction rules of the template libraries used for report generation.

Library	Source reports	Entries	Construction rules
SMU-HMC general sentences	7,147 SMU-HMC reports (completeness score ≥0.9; test patients excluded)	400 (gland 358, measurement 24, lymph node 18)	Gland 1,793 → 400 most frequent → 358 after a manual exclusion list of 42; measurement 128 → 40 most frequent → 24 after a manual exclusion list of 15 and removal of 1 incomplete sentence; lymph node 514 → 18 (frequency ≥20)
SMU-HMC nodule templates	7,147 SMU-HMC reports (completeness score ≥0.9; test patients excluded)	204	7,478 nodule sentences → 4,656 templates → frequency ≥3: 233 → 29 with fixed echogenicity wording removed
SMU-HMC impression clauses	7,147 SMU-HMC reports (completeness score ≥0.9; test patients excluded)	1,076	Frequency ≥2
SMU-HMC whole impressions	7,147 SMU-HMC reports (completeness score ≥0.9; test patients excluded)	3,094	Identical impressions merged
KMVE library (revised)	1,313 reports of the official KMVE training split (406 near-duplicates of test reports excluded)	82 nodule templates, 259 gland sentences in 23 groups, 91 lymph-node sentences, 10 gland-size openers	Nodule sentences converted into slot templates (location, number, echogenicity, margin, shape, echogenic foci, vascularity) and merged with frequencies; gland sentences grouped by gland size, parenchymal echo and parenchymal vascularity; slots filled from the tool outputs; sizes masked as in KMVE reports

Slots in the SMU-HMC templates (entries containing the slot): location (144); size (142); sizes of several nodules (51); composition (23); echogenicity (172); margin (192); shape (43); aspect ratio (19); internal echo (63); echogenic foci (37); vascularity (197); TI-RADS category (3,528).

Supplementary Table S25. Templates removed by manual review from the 400 most frequent gland templates and the 40 most frequent measurement templates of the SMU-HMC library.

Library	Reason for removal	Occurrences	Template (original)	Template (English translation)
Gland	Postoperative or residual gland	32	甲状腺术后。	Post-thyroidectomy.
Gland	Postoperative or residual gland	16	甲状腺部分切除术后。	Post partial thyroidectomy.
Gland	Postoperative or residual gland	5	甲状腺术后：左侧残留甲状腺大小约<SIZE3>，右侧残留甲状腺大小约 <SIZE3>。	Post-thyroidectomy: residual left thyroid about <SIZE3>, residual right thyroid about <SIZE3>.
Gland	Postoperative or residual gland	4	甲状腺术后：甲状腺左叶大小:<SIZE3>，右叶大小:<SIZE3>。	Post-thyroidectomy: thyroid left lobe: <SIZE3>, right lobe: <SIZE3>.
Gland	Postoperative or residual gland	3	左侧残留甲状腺大小:<SIZE3>，右侧残留甲状腺大小:<SIZE3>。	Residual left thyroid: <SIZE3>, residual right thyroid: <SIZE3>.
Gland	Postoperative or residual gland	2	甲状腺术后：左侧残留甲状腺大小:<SIZE3>，右侧残留甲状腺大小:<SIZE3>，峡部厚<NUM>mm。	Post-thyroidectomy: residual left thyroid: <SIZE3>, residual right thyroid: <SIZE3>, isthmus <NUM> mm thick.
Gland	Postoperative or residual gland	2	甲状腺部分切除术后：甲状腺左叶大小:<SIZE3>，右叶大小:<SIZE3>。	Post partial thyroidectomy: thyroid left lobe: <SIZE3>, right lobe: <SIZE3>.
Gland	Postoperative or residual gland	2	左侧残留甲状腺大小约<SIZE3>，右侧残留甲状腺大小约 <SIZE3>。	Residual left thyroid about <SIZE3>, residual right thyroid about <SIZE3>.
Gland	Postoperative or residual gland	2	甲状腺 CA 术后。	Post-thyroidectomy for thyroid cancer.
Gland	Measurement sentence, not a gland description	252	甲状腺左叶 <SIZE3>，右叶 <SIZE3>，峡部<NUM>mm。	Thyroid left lobe <SIZE3>, right lobe <SIZE3>, isthmus <NUM> mm.
Gland	Measurement sentence, not a gland description	40	甲状腺左叶大小约<NUM>mm×<NUM>mm×<NUM>mm，右叶大小约 <NUM>mm×<NUM>mm×<NUM>mm，峡部厚约 <NUM>mm。	Thyroid left lobe about <NUM> mm × <NUM> mm × <NUM> mm, right lobe about <NUM> mm × <NUM> mm × <NUM> mm, isthmus about <NUM> mm thick.
Gland	Measurement sentence, not a gland description	37	甲状腺左叶大小:<SIZE2>，右叶大小:<SIZE2>，峡部厚<NUM>mm。	Thyroid left lobe: <SIZE2>, right lobe: <SIZE2>, isthmus <NUM> mm thick.
Gland	Measurement sentence, not a gland description	19	甲状腺左叶大小:<SIZE3>，右叶大小:<SIZE3>。	Thyroid left lobe: <SIZE3>, right lobe: <SIZE3>.

(continued)

Library	Reason for removal	Occurrences	Template (original)	Template (English translation)
Gland	Measurement sentence, not a gland description	9	甲状腺左叶约 <SIZE3>，右叶大小约 <SIZE3>，峡部约 <NUM>mm。	Thyroid left lobe about <SIZE3>, right lobe about <SIZE3>, isthmus about <NUM> mm.
Gland	Measurement sentence, not a gland description	8	甲状腺左侧叶大小:<SIZE3>，右侧叶大小:<SIZE3>，峡部厚 <NUM>mm。	Thyroid left lobe: <SIZE3>, right lobe: <SIZE3>, isthmus <NUM> mm thick.
Gland	Measurement sentence, not a gland description	4	甲状腺左叶 <SIZE3>，右叶 <SIZE3>，峡部 mm。	Thyroid left lobe <SIZE3>, right lobe <SIZE3>, isthmus mm.
Gland	Measurement sentence, not a gland description	4	甲状腺右叶大小:<SIZE3>，左叶大小:<SIZE3>。	Thyroid right lobe: <SIZE3>, left lobe: <SIZE3>.
Gland	Measurement sentence, not a gland description	2	甲状腺左叶大小约 <SIZE3>，右叶大小约 <SIZE3>。	Thyroid left lobe about <SIZE3>, right lobe about <SIZE3>.
Gland	Glandular vascularity only	107	CDFI：腺体内见少量纤细分枝状彩色血流信号。	CDFI: a small amount of fine branching colour flow signal in the gland.
Gland	Glandular vascularity only	26	CDFI：腺体内可探及较丰富血流信号。	CDFI: relatively abundant flow signal in the gland.
Gland	Glandular vascularity only	13	CDFI：腺体内见血流信号轻度增多，呈点状或分枝状。	CDFI: mildly increased flow signal in the gland, dotted or branching.
Gland	Glandular vascularity only	7	CDFI：腺体内见血管轻度增多，呈点状或分枝状。	CDFI: mildly increased vessels in the gland, dotted or branching.
Gland	Glandular vascularity only	3	CDFI：腺体内血管弥漫性明显增多。	CDFI: diffusely and markedly increased vessels in the gland.
Gland	Glandular vascularity only	3	CDFI 于甲状腺腺体内可见极丰富血流信号显示，呈 “ 火海样 ”。	CDFI: very abundant flow signal in the thyroid gland, with a “thyroid inferno” pattern.
Gland	Glandular vascularity only	3	腺体内血管弥漫性明显增多。	Diffusely and markedly increased vessels in the gland.
Gland	Glandular vascularity only	2	CDFI：腺体内可见较丰富血流信号显示，呈 “ 火海样 ”。	CDFI: relatively abundant flow signal in the gland, with a “thyroid inferno” pattern.
Gland	Glandular vascularity only	2	CDFI：腺体内血流弥漫性明显增多，呈火海样。	CDFI: diffusely and markedly increased flow in the gland, with a thyroid inferno pattern.
Gland	Glandular vascularity only	2	CDFI：腺体内血流信号弥漫性明显增多。	CDFI: diffusely and markedly increased flow signal in the gland.
Gland	Glandular vascularity only	2	CDFI：腺体内血流信号稍丰富。	CDFI: slightly abundant flow signal in the gland.
Gland	Glandular vascularity only	2	CDFI：腺体内可探及丰富血流信号。	CDFI: abundant flow signal in the gland.
Gland	Glandular vascularity only	2	CDFI：腺体内血流信号弥漫性增多。	CDFI: diffusely increased flow signal in the gland.

(continued)

Library	Reason for removal	Occurrences	Template (original)	Template (English translation)
Gland	Focal lesion, not a gland description	10	双侧甲状腺内可见多个无回声，大小约<SIZE2>、<SIZE2>、<SIZE2>等，内均可见一强回声光点，CDFI：其内未见明确血流信号。	Multiple anechoic lesions in both lobes, about <SIZE2>, <SIZE2>, <SIZE2> and others, each containing an echogenic dot; CDFI: no definite flow signal within.
Gland	Focal lesion, not a gland description	6	甲状腺双侧叶见多个无回声，较大约<SIZE2>，边界清，暗区内见点状强回声，CDFI：其内未见血流信号。	Multiple anechoic lesions in both lobes, the larger about <SIZE2>, well defined, with punctate echogenic foci in the anechoic area; CDFI: no flow signal within.
Gland	Focal lesion, not a gland description	4	双侧甲状腺内可见多个无回声，大小约<SIZE2>、<SIZE2>、<SIZE2>，内均可见一强回声光点，CDFI：其内未见明确血流信号。	Multiple anechoic lesions in both lobes, about <SIZE2>, <SIZE2> and <SIZE2>, each containing an echogenic dot; CDFI: no definite flow signal within.
Gland	Focal lesion, not a gland description	4	甲状腺右侧叶见多个无回声，较大约<SIZE2>，边界清，暗区内见点状强回声，CDFI：其内未见血流信号。	Multiple anechoic lesions in the right lobe, the larger about <SIZE2>, well defined, with punctate echogenic foci in the anechoic area; CDFI: no flow signal within.
Gland	Focal lesion, not a gland description	4	双侧甲状腺内可见多个无回声，大小约<SIZE2>、<SIZE2>，内均可见一强回声光点，CDFI：其内未见明确血流信号。	Multiple anechoic lesions in both lobes, about <SIZE2> and <SIZE2>, each containing an echogenic dot; CDFI: no definite flow signal within.
Gland	Focal lesion, not a gland description	2	甲状腺左叶另见强回声，大小约<SIZE2>。	Another echogenic focus in the left lobe, about <SIZE2>.
Gland	Focal lesion, not a gland description	2	双侧甲状腺见无回声，左侧较大者约<SIZE2>，右侧较大者约<SIZE2>，边界清，CDFI：其内未见血流信号。	Anechoic lesions in both lobes, the larger on the left about <SIZE2> and on the right about <SIZE2>, well defined; CDFI: no flow signal within.
Gland	Focal lesion, not a gland description	2	甲状腺右叶见实质性光团，大小约<SIZE2>，CDFI：其内未见血流信号。	A solid mass in the right lobe, about <SIZE2>; CDFI: no flow signal within.
Gland	Focal lesion, not a gland description	2	甲状腺右叶探及实质性光团，大小约<SIZE2>、<SIZE2>，CDFI：其内未探及血流信号。	Solid masses in the right lobe, about <SIZE2> and <SIZE2>; CDFI: no flow signal detected within.
Gland	Focal lesion, not a gland description	2	甲状腺左叶中部见强回声，大小约<SIZE2>。	An echogenic focus in the middle of the left lobe, about <SIZE2>.

(continued)

Library	Reason for removal	Occurrences	Template (original)	Template (English translation)
Gland	Focal lesion, not a gland description	2	甲状腺右叶中部见强回声后伴声影，大小约<SIZE2>。	An echogenic focus with posterior acoustic shadowing in the middle of the right lobe, about <SIZE2>.
Measurement	Emergency or bedside examination note	17	急诊超声检查：甲状腺左叶大小约<SIZE3>，右叶大小约<SIZE3>，峡部厚约<NUM>mm。	Emergency ultrasound: thyroid left lobe about <SIZE3>, right lobe about <SIZE3>, isthmus about <NUM> mm thick.
Measurement	Emergency or bedside examination note	14	急诊超声检查：甲状腺左叶大小:<SIZE3>，右叶大小:<SIZE3>，峡部厚<NUM>mm。	Emergency ultrasound: thyroid left lobe: <SIZE3>, right lobe: <SIZE3>, isthmus <NUM> mm thick.
Measurement	Emergency or bedside examination note	11	床旁超声检查，图像未能留存，结果仅供参考：甲状腺左叶大小约<SIZE3>，右叶大小约<SIZE3>，峡部厚约<NUM>mm。	Bedside ultrasound, images not stored, results for reference only: thyroid left lobe about <SIZE3>, right lobe about <SIZE3>, isthmus about <NUM> mm thick.
Measurement	Emergency or bedside examination note	6	急诊彩超：甲状腺左叶大小:<SIZE3>，右叶大小:<SIZE3>，峡部厚<NUM>mm。	Emergency colour Doppler ultrasound: thyroid left lobe: <SIZE3>, right lobe: <SIZE3>, isthmus <NUM> mm thick.
Measurement	Emergency or bedside examination note	4	床旁超声检查，图像未能留存，结果仅供参考：甲状腺左叶大小:<SIZE3>，右叶大小:<SIZE3>，峡部厚<NUM>mm。	Bedside ultrasound, images not stored, results for reference only: thyroid left lobe: <SIZE3>, right lobe: <SIZE3>, isthmus <NUM> mm thick.
Measurement	Emergency or bedside examination note	3	急诊超声检查：甲状腺右叶大小:<SIZE3>，左叶大小:<SIZE3>，峡部厚<NUM>mm。	Emergency ultrasound: thyroid right lobe: <SIZE3>, left lobe: <SIZE3>, isthmus <NUM> mm thick.
Measurement	Emergency or bedside examination note	2	急诊超声检查：甲状腺左叶大小约<SIZE3>，右叶大小约<SIZE3>，峡部<NUM>mm。	Emergency ultrasound: thyroid left lobe about <SIZE3>, right lobe about <SIZE3>, isthmus <NUM> mm.
Measurement	Postoperative examination	8	甲状腺部分切除术后：甲状腺左叶大小:<SIZE3>，右叶大小:<SIZE3>，峡部厚<NUM>mm。	Post partial thyroidectomy: thyroid left lobe: <SIZE3>, right lobe: <SIZE3>, isthmus <NUM> mm thick.
Measurement	Postoperative examination	4	甲状腺术后：甲状腺左叶大小:<SIZE3>，右叶大小:<SIZE3>，峡部厚<NUM>mm。	Post-thyroidectomy: thyroid left lobe: <SIZE3>, right lobe: <SIZE3>, isthmus <NUM> mm thick.

(continued)

Library	Reason for removal	Occurrences	Template (original)	Template (English translation)
Measurement	Postoperative examination	3	甲状腺术后：甲状腺左叶大小约 <SIZE3>，右叶大小约 <SIZE3>，峡部厚约 <NUM> mm。	Post-thyroidectomy: thyroid left lobe about <SIZE3>, right lobe about <SIZE3>, isthmus about <NUM> mm thick.
Measurement	Postoperative examination	2	甲状腺部分切除术后：甲状腺右叶大小:<SIZE3>，左叶大小:<SIZE3>，峡部厚 <NUM>mm。	Post partial thyroidectomy: thyroid right lobe: <SIZE3>, left lobe: <SIZE3>, isthmus <NUM> mm thick.
Measurement	Missing value	16	甲状腺左叶大小:<SIZE3>，右叶大小:<SIZE3>，峡部厚 mm。	Thyroid left lobe: <SIZE3>, right lobe: <SIZE3>, isthmus mm thick.
Measurement	Missing value	3	甲状腺左叶大小约 <SIZE3>，右叶大小约 <SIZE3>，峡部厚 mm。	Thyroid left lobe about <SIZE3>, right lobe about <SIZE3>, isthmus mm thick.
Measurement	Missing value	2	甲状腺左叶大小:× <SIZE2>，右叶大小:<SIZE3>，峡部厚 <NUM>mm。	Thyroid left lobe: × <SIZE2>, right lobe: <SIZE3>, isthmus <NUM> mm thick.
Measurement	Measurement noted as unsatisfactory	1	（甲状腺体积过大，测量欠满意）甲状腺左叶大小:<SIZE3>，右叶大小:<SIZE3>，峡部厚 <NUM>mm。	(Thyroid too large; measurement unsatisfactory) Thyroid left lobe: <SIZE3>, right lobe: <SIZE3>, isthmus <NUM> mm thick.

Templates keep their slot placeholders: <SIZE2>, two-dimensional size; <SIZE3>, three-dimensional size; <NUM>, number. Occurrences are the numbers of source clauses mapped to each template.

Supplementary Table S26. Inputs received by each method in the report-generation comparison.

Method	Images per case: SMU-HMC	Images per case: ZJH-TS	Images per case: KMVE	Other input	Report written by
ThyroidXAgent	all (median 13; 4–65)	all (median 11.5; 4–46)	2	Structured outputs of segmentation, classification, measurement, lymph-node screening and fusion tools	Template retrieval (SMU-HMC library; KMVE library for KMVE)
GPT-4o, GPT-5, Gemini 2.5 Pro, Qwen3.5 Plus, Claude Sonnet 4.6 (API)	up to 5, random	up to 5, random	2	Prompt with the report structure and 3 example reports in the format of the dataset (few-shot)	The model
MedGemma (local)	up to 5, random	up to 5, random	2	Same prompt as above	The model

(continued)

Method	Images per case: SMU-HMC	Images per case: ZJH-TS	Images per case: KMVE	Other input	Report written by
LLaVA-Med (local)	2, random	2, random	2	English version of the prompt	The model, in English; translated into Chinese with GPT-4o-mini
KMVE model (trained on KMVE)	2, random	2, random	not evaluated	None	The model
Static pipeline	all	all	2	Same tools as ThyroidXAgent, called by fixed rules	Same templates as ThyroidXAgent

Random image selection used seed 42. Image counts for the baselines are those set in the released generation scripts; for the KMVE model on ZJH-TS they were confirmed from the saved outputs.